

Nassau Candy Factory Reallocation Optimization

Intelligent Logistics Assignment Engine — Research Paper & Analytical Methodology

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Abstract

This paper details the development of an intelligent factory-product reassignment engine for Nassau Candy Distributor. By transitioning from static assignment rules to predictive analytics, the system enables real-time optimisation of logistics operations, directly mitigating suboptimal shipping distances, elevated lead times, and progressive margin erosion. The methodology encompasses synthetic data generation, multi-model machine learning evaluation, route clustering, and a scenario-based simulation engine capable of quantifying profit impact across alternative factory allocations.

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Data Preparation & Exploratory Data Analysis

A synthetic dataset comprising **5,000 logistics and sales records** was generated to faithfully replicate the operational characteristics of Nassau Candy Distributor. The dataset spans multiple factories, product lines, shipping modes, and regional destinations, providing sufficient variance for robust model training and evaluation.

Key Preprocessing Steps

- **Feature Engineering:** The target variable *Lead Time (Days)* was derived as the arithmetic difference between Ship Date and Order Date, establishing a consistent prediction objective across all records.
- **Encoding:** One-Hot Encoding was applied to all categorical variables — including Region, Product Name, Origin Factory, and Ship Mode — to produce a fully numerical feature matrix suitable for

machine learning algorithms.

■ **Normalisation:** Continuous numerical features such as Cost and Sales were min-max scaled to ensure consistent magnitude and prevent dominance by high-range variables during model training.

Exploratory Data Analysis (EDA) revealed that specific factory-region routes exhibit high variance in lead times, statistically confirming the operational inefficiencies inherent in static factory assignment. Certain origin-destination pairs showed lead time standard deviations exceeding three times the global mean — a key motivator for the predictive reassignment approach.

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Predictive Modelling Objective

The primary modelling objective is to **predict expected shipping lead time** given a combination of Origin Factory, Destination Region, Product, and Ship Mode. Accurate lead-time prediction is the foundation upon which the simulation engine evaluates alternative factory assignments.

Candidate Models

Model	Strengths	Role
Linear Regression	Simple, interpretable	Baseline benchmark
Random Forest	Non-linear, robust to noise	Primary candidate
Gradient Boosting	Sequential error minimisation	High-accuracy challenger

Evaluation Metrics

Each candidate model was assessed against three complementary metrics to capture both average performance and sensitivity to outlying predictions:

Metric	Description	Purpose
RMSE	Root Mean Square Error	Penalises large prediction errors heavily
MAE	Mean Absolute Error	Captures average prediction divergence
R ²	Coefficient of Determination	Measures variance explained by the model

The best-performing model — selected on the basis of lowest RMSE and highest R² on the held-out test set — was serialised and integrated directly into the simulation engine, ensuring production inference is consistent with the evaluated configuration.

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Route & Product Clustering

To supplement the regression model with structural insight, an unsupervised clustering layer was applied to all observed factory-region-product routes. Routes were grouped according to performance similarity, using average lead time as the primary clustering signal.

This analysis produced two operationally significant outputs:

- **Congested Combinations:** Region-product pairs where demand concentration and limited factory coverage systematically inflate lead times, identifying high-priority reallocation candidates.
- **Persistently Slow Routes:** Factory-region lanes that consistently underperform regardless of product mix or season, flagged within the system dashboard for proactive managerial review.

Cluster membership is stored alongside each route's historical record, enabling the simulation engine to surface cluster-level risk signals alongside individual lead-time predictions.

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Scenario Simulation Engine

The Scenario Simulation Engine represents the core innovation of this project. It operationalises predictive modelling by translating model outputs into actionable reallocation recommendations with quantified financial impact.

Engine Workflow — Four-Stage Pipeline

◆ Stage 1 — Baseline Prediction

For any given product-route combination, the engine invokes the trained model to predict the current expected lead time under the existing factory assignment.

◆ Stage 2 — Alternate Factory Simulation

The engine systematically simulates reassignment to every eligible alternative factory, generating a predicted lead time for each candidate configuration.

◆ Stage 3 — Profit Impact Estimation

Predicted lead-time deltas are translated into estimated profit impact, operating under the empirically supported assumption that delivery speed improvements correlate positively with margin outcomes (reduced spoilage, higher fill rates, fewer expedite surcharges).

◆ Stage 4 — Confidence Scoring

Each recommendation is accompanied by a Confidence Score derived from the volume of historical data available for the simulated route — higher data density yields greater confidence, enabling planners to calibrate risk accordingly.

The engine's output is surfaced through an interactive decision dashboard, allowing operations managers to review ranked reallocation options, filter by confidence threshold, and approve changes within a single workflow.

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Conclusion

By unifying predictive machine learning with scenario-based optimisation logic, the Nassau Candy Factory Reallocation System delivers a significant capability uplift over the incumbent static assignment framework. The system empowers operations teams to make **data-driven logistics decisions in real time**, improving shipping efficiency without sacrificing product quality or profitability.

Anticipated operational benefits include measurable reductions in average lead time, decreased exposure to high-variance routes, and enhanced margin performance through improved delivery reliability. Future development will focus on integrating live ERP data feeds, expanding the simulation engine to incorporate carrier capacity constraints, and deploying reinforcement learning to enable continuous self-optimisation of factory-product assignments.